# Environmental Factors that Impact College Admission

TEAM 2152-901

TSA CONFERENCE 2023

## INTRODUCTION

*It is common knowledge that college in America is difficult to get into, as high school students spend the majority of their time trying to build up their resume in order to better their chances of getting into a prestigious institution. However this issue raises many questions regarding how much of the college admissions process do students actually have control over. An* ***analysis of the environmental factors surrounding college admissions*** *is crucial in understanding the ethics of college admissions as well as how much control students even have over their chance of admission.*

## PURPOSE

*The research in this project was focused on the major environmental factors that impact college admissions in order* ***to learn whether or not the college admissions process as it stands is even ethical and to predict what the future of college admissions will look like if the current trends continue****. By understanding this we can further the conversation on the power these large institutions have as they design these admission systems to serve themselves.*

## METHODS

*Data obtention and analysis through the use of* ***documents and records*** *(Stanford, Harvard, UPenn),* ***correlation research****, data cleaning, and* ***machine learning*** *(AWS Sagemaker and DataRobot platforms).*

## RESULTS

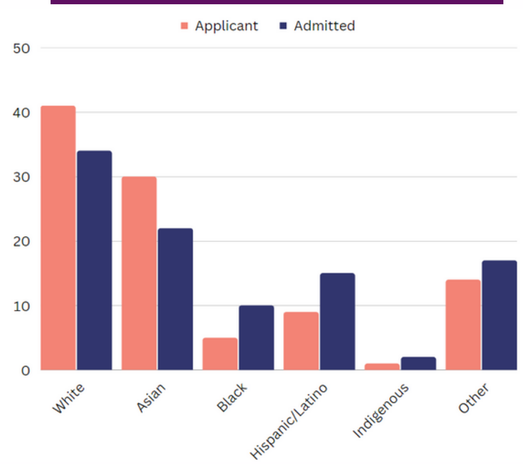



**Figure 1.** *Average Demographics of the Applicant Pool vs. Admitted Class*, Fall 2022

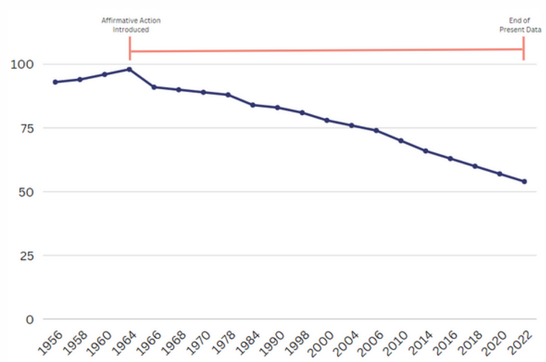



**Figure 2.** *Percentage of White Students Admitted to an American University*, 1956-2022

## RESULTS

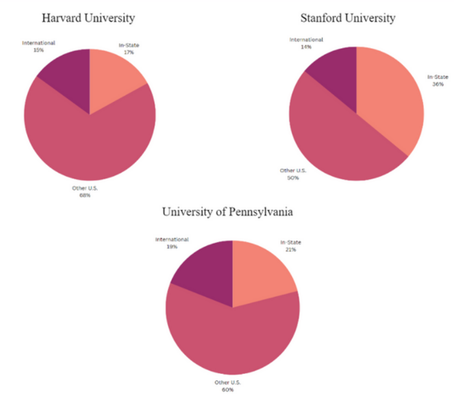



**Figure 4.** *Geographic Diversity of Accepted Undergraduate Class*, Fall 2022

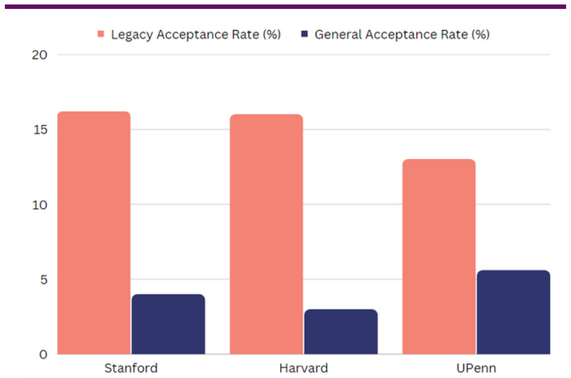
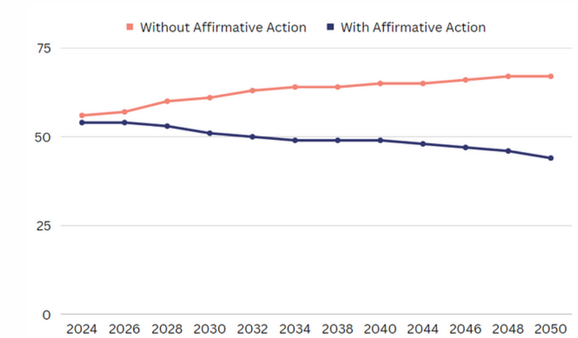



**Figure 5.** Comparison of *Legacy Student vs General Acceptance Rate*, Fall 2022

**Figure 3.** Predictions Made by the Model *With and Without Affirmative Action*, 2024-2050




## CONCLUSIONS

**Race:** The racial demographics changed from the applicant pool to admitted class, and affirmative action shifted the trend of percent white.

**Location:** The universities had similar geographic makeup, with U.S. and instate being the largest.

**Status as a Legacy:** While legacies have a higher acceptance rate, they do not make up a majority of the admitted class.

## NEXT STEPS

*Analyze the trend with and without affirmative action for* ***individual racial groups****, compare the applicant pool and admitted class of* ***other universities****, research the* ***events*** *that could change the geographical and legacy demographics, and find the* ***correlation between the factors*** *to have more context and end the ethics debate.*

## REFERENCES

***Official Stanford, Harvard, and UPenn Websites:*** *demographics and acceptance rates*

***U.S. National Center for Education Statistics:*** *percent white in U.S. Universities since 1956*